Code No: 136AG **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is a state-space search in problem-solving?	[2]
b)	Differentiate between exhaustive searches and heuristic searches.	[3]
c)	What do you mean by semantic network?	[2]
d)	Define predicate logic.	[3]
e)	What is a truth maintenance system?	[2]
f)	Describe the applications of expert system.	[3]
g)	Define inductive learning.	[2]
h)	Define recurrent networks.	[3]
i)	What are grammars in NLP?	[2]
j)	What is semantic analysis in NLP?	[3]

PART – B

(50 Marks)

2. Use the A* algorithm to find the shortest path in a given grid-based maze. Describe the heuristic used and show the path found by the algorithm. [10]

OR

3. Analyze a simple game tree (e.g., Tic-Tac-Toe) and demonstrate how Alpha-Beta pruning reduces the number of nodes evaluated compared to the basic Minimax algorithm. Explain each step with a small example. [10]

4. Create a simple Prolog program to solve the classic "parent" relationship problem. Define the relationships and provide sample queries and their expected outcomes. [10]

OR

5.a) Determine whether the following statement is Satisfiable, Contradictory, or Valid: $(P \wedge Q) \wedge \neg(P \wedge Q)$

b) Determine whether the following statement is Satisfiable, Contradictory, or Valid: $(P \vee \neg P) \wedge \neg(P \vee \neg P)$ [5+5]

6. Write a detailed note on Bayesian Belief Networks. [10]

OR

7. Apply the phases of building an expert system to design a simple expert system for diagnosing plant diseases. Describe each phase in detail. [10]

8. Analyze the results of applying k-means clustering to a simple dataset (e.g., clustering points into two groups). [10]

OR

9. Create a basic decision tree model to classify types of fruits based on their features (e.g., color, size, shape). Provide the decision tree and describe how it classifies a new fruit. [10]

10. Apply the concept of case grammars to parse the following sentence: "The dog chased the cat." Identify the roles (e.g., agent, object) for each word in the sentence. [10]

OR

11. Evaluate the effectiveness of semantic analysis in resolving ambiguities in natural language processing. Provide examples where semantic analysis successfully disambiguates a sentence. [10]

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